

Impact Statement Card

SLM : Agriculture and Climate

Project	Domain-Specific SLM for Agriculture and Climate
Team	Kulal
Team Members	Adepitan Oluwatosin, Samuel Ekaette
Date	September 2026
Version	v1.0

This Impact Statement Card describes the intended benefits, potential risks, and accountability commitments of the SLM : Agriculture and Climate project. The model is a Small Language Model fine-tuned to answer agricultural and climate questions for smallholder farmers in Nigeria. It is built on FAO and CGIAR data, uses FAISS-based semantic retrieval to ground responses in verified source material, and is designed to work in low-bandwidth rural environments.

1. Model Purpose and Intended Use

Agriculture employs over 70% of Nigeria's rural population, yet farmers face growing challenges from unpredictable rainfall, shifting planting seasons, and novel pest outbreaks driven by climate change. Existing tools are English-only, assume good internet connectivity, and offer no hyperlocal climate context. This model is built to close that gap.

What the model does

- Takes a plain-language question from a smallholder farmer as input.
- Retrieves the most relevant documents from a curated agricultural and climate knowledge base using FAISS semantic search.
- Generates a grounded, locally relevant advisory response covering planting schedules, pest and disease management, and climate adaptation.

Who it is for

- Primary users: Smallholder farmers in Nigeria, particularly in the Sudan Savanna, Guinea Savanna, Derived Savanna, and Rainforest Belt zones.
- Secondary users: Agricultural extension officers, NGOs supporting rural farmers, and researchers studying AI for low-resource agricultural contexts.

2. Intended Benefits

Benefit	Who It Helps	How
Better planting decisions	Smallholder farmers	The model provides season and zone-specific guidance on when to plant, what to plant, and how to prepare soil, reducing reliance on guesswork or outdated advice.

Faster pest and disease response	Farmers, extension officers	Farmers can describe a symptom and get practical advice quickly, without waiting days for an extension officer visit that may never come to remote areas.
Climate adaptation support	Farmers in climate-vulnerable zones	The model draws on CGIAR and FAO climate data to help farmers adjust practices as rainfall patterns shift, supporting food security in the most vulnerable communities.
Accessible information	Low-literacy and low-connectivity users	Responses are written at a Grade 8 reading level or below. The model is designed to run in low-bandwidth environments, making it usable on basic smartphones without reliable internet.
Extension service reach	Government and NGO extension programmes	The model can act as a first-line advisory tool, freeing extension officers to focus on complex cases and reaching communities that current services do not cover.
Knowledge preservation	Farming communities	Indigenous farming knowledge gathered during community interviews is incorporated into the training data, preserving and validating local practices alongside formal agronomic advice.

3. Potential Risks and Harms

Any AI system giving advice that affects livelihoods and food supply carries real risk. The following table documents the risks the team has identified and the steps taken to reduce them.

Risk	Likelihood	Potential Harm	Mitigation
Factually incorrect advice	Medium	A farmer follows wrong guidance on planting timing or pest treatment, leading to crop loss.	All Q&A; pairs are reviewed by a certified agronomist before inclusion. Model outputs are evaluated against a held-out test set reviewed by agronomists before deployment.
Context mismatch	Medium	Advice designed for one agroecological zone is applied in another where conditions differ.	Every record is tagged with zone, season, and crop. The retrieval system prioritises zone-matched documents. Users are prompted to confirm their location context.
Over-reliance on the model	Medium	Farmers stop consulting extension officers entirely and trust the model for decisions it cannot safely make alone.	Responses include explicit prompts to consult a local extension officer for high-stakes decisions. The model is positioned as a first-line tool, not a replacement for professional advice.
Language and literacy gaps	High (near-term)	Farmers who are not comfortable reading English miss out on the tool entirely.	English-first design is a starting point, not an end point. Yoruba, Hausa, and Igbo extension is planned. Audio interface options are under consideration for future phases.

Outdated information	Low to medium	Seasonal guidance or pest alerts that were accurate when collected become misleading as climate patterns shift.	The dataset is updated twice yearly aligned with the Nigerian farming calendar. A feedback mechanism allows extension officers to flag outdated advice.
Amplifying existing bias	Medium	If training data underrepresents women farmers or southern zones, advice for those groups will be weaker.	Gender and geographic representation targets are enforced in data collection. At least 50% of farmer interview participants are female. Zone representation is monitored after each collection phase.
Misuse outside intended scope	Low	The model is used for general-purpose tasks it was not designed for, producing unreliable outputs.	Out-of-scope uses are clearly defined in the Data Card and model documentation. The model is released under a research-only licence.

4. Affected Communities

Directly affected

- Smallholder farmers in Nigeria, particularly women who carry the majority of agricultural labour and are underserved by current digital tools.
- Farming communities in climate-vulnerable agroecological zones, including the Sudan Savanna in the north where drought risk is highest.
- Agricultural extension officers whose workload and reach the model is designed to complement.

Indirectly affected

- Rural households that depend on smallholder farm income for food security and school fees.
- Local food markets and supply chains that depend on smallholder output.
- Nigerian government agricultural programmes whose effectiveness the model could support or undermine depending on how it is deployed.

Communities at risk of exclusion

- Farmers who speak only Yoruba, Hausa, or Igbo and have limited English literacy. These users are the most important long-term target but are underserved in the current version.
- Farmers with no smartphone access at all. The model requires at minimum a basic smartphone, which not all rural farmers in Nigeria currently have.

5. Accountability and Safeguards

Agronomist review	All training data is reviewed by a certified agronomist before the model sees it. Model outputs on the evaluation set are reviewed by an agronomist before any deployment decision.
Feedback mechanism	Extension officers and NGO partners can flag incorrect or harmful advice through a structured feedback form. Flagged responses are reviewed within 30 days.

Community reporting	Participating farming communities are given a plain-language summary of what the model was trained on and how to raise concerns.
Version control	Every model version is tracked alongside the dataset version it was trained on. Rollback is possible if a new version performs worse on safety-critical queries.
No silent deletion	Training records found to contain errors are deprecated and flagged in the dataset rather than silently removed, to maintain a full audit trail.
Scope enforcement	The model documentation explicitly states it is not a substitute for certified professional agronomic advice. High-stakes queries trigger a prompt to contact a local extension officer.
Biannual updates	The dataset and model are reviewed and updated twice yearly, aligned with the Nigerian farming calendar, to keep seasonal guidance accurate.
Five-year review	A full community re-consultation will be conducted five years after release to assess whether the model is still serving the communities it was built for.

6. Alignment with Project Values

The impact commitments above trace directly back to the values that shaped the problem statement. Each value is reflected in a concrete safeguard or design decision, not just a stated intention.

Value	Impact Commitment
Equity and Inclusion	Gender and geographic representation targets are enforced in data collection. Multilingual extension to Yoruba, Hausa, and Igbo is a named priority for future phases.
Community Agency	Farmers contributed knowledge as active participants, not just data sources. Findings are shared back to communities. Communities can raise concerns through NGO partners.
Relevance	Every advisory response is grounded in zone, season, and crop-specific source material. General-purpose advice is explicitly out of scope.
Accessibility	Grade 8 or below reading standard. Low-bandwidth deployment design. Audio interface under consideration for future phases.
Transparency	Full provenance on every training record. This Impact Statement Card and the Data Card are both publicly available in the project repository.
Avoiding Harm	Agronomist review at every stage. Risk table documented and publicly available. Model explicitly scoped away from replacing professional advice.